Model of Quantum Computation

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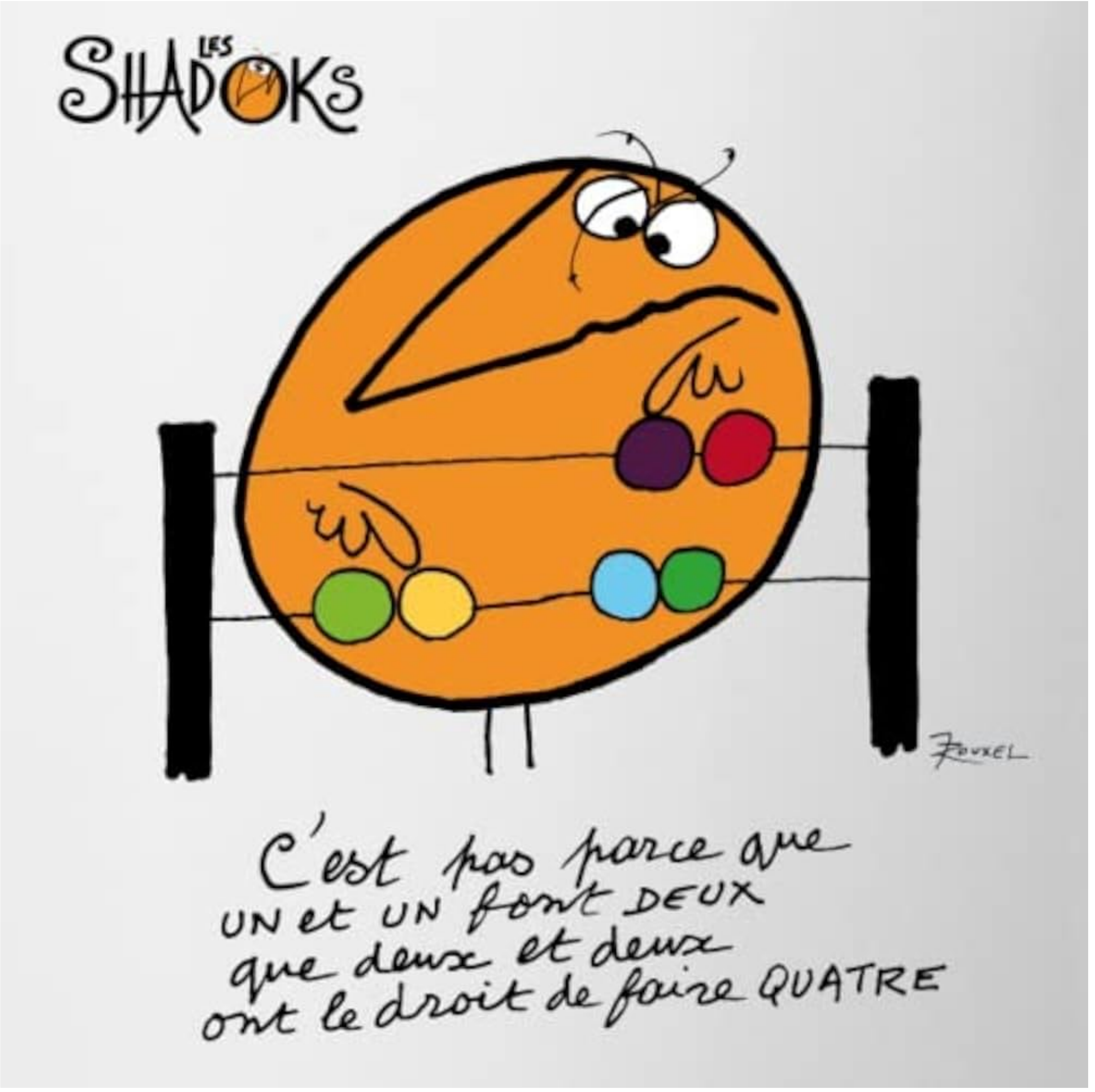
Outline

- Introduction
- Classical Computing Model
- Quantum Computing Model
- Open Questions



Abstracting Computation

1. How to encode Information ? (State, Symbol, etc…)
2. How we modify information ? (Transition, Group Action, etc…)
3. How to get a result ? (Halting, Output, etc…)

Example, Tag rewriting system

- Alphabet ($\Sigma = \{A_1, A_2, \cdots, A_n\}$)
- Input/Output : word $\underline{A_i A_j A_k \cdots}$
- Transition $\delta : \Sigma \to W(\Sigma)$
- Evolution : $\underline{A_i A_j W} \mapsto \underline{W\delta(A_i)}$

Turing Machine -Definition

- $TM = \langle \Sigma, Q, q_0, \delta, F \rangle$

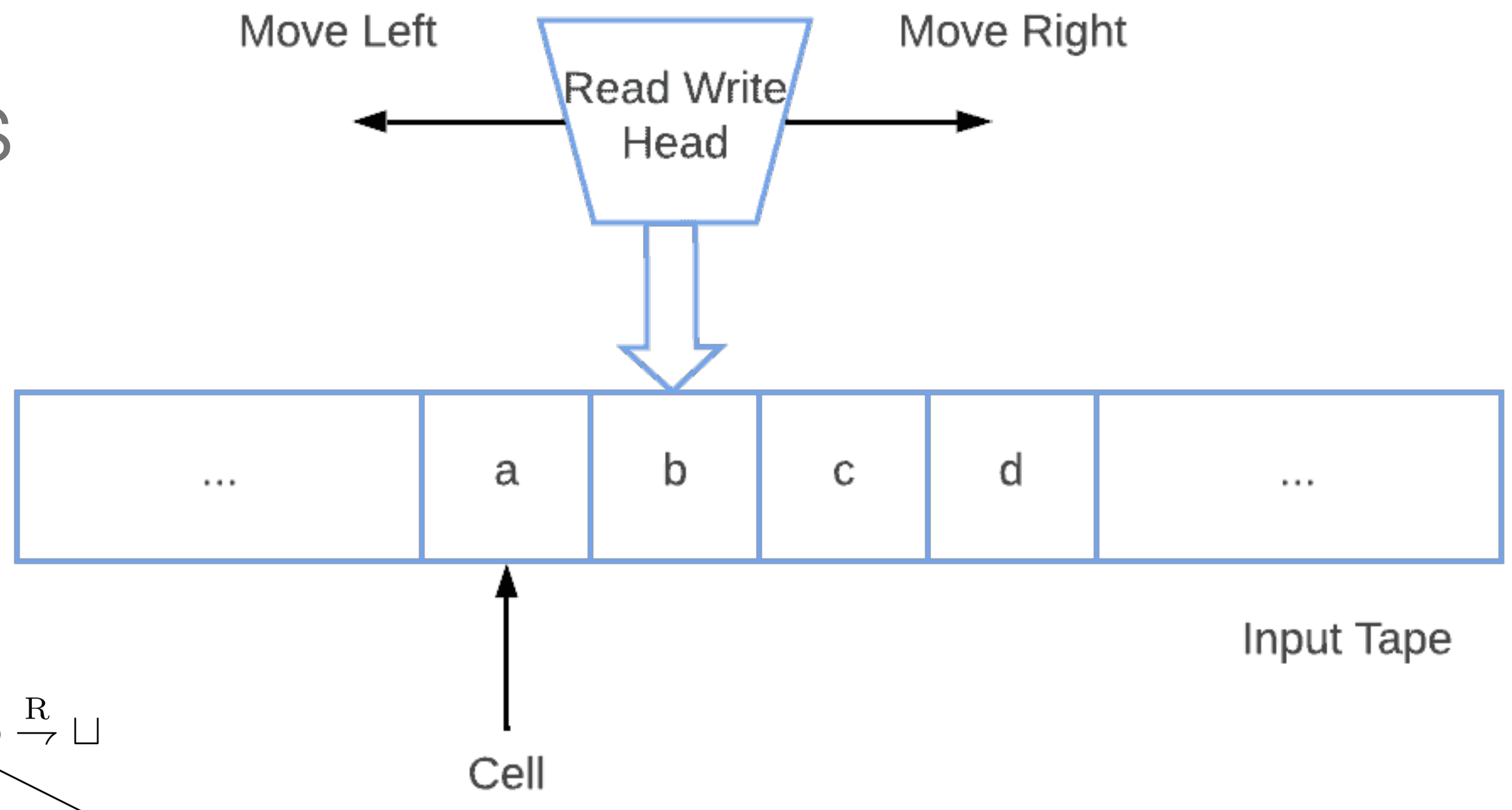
Final States

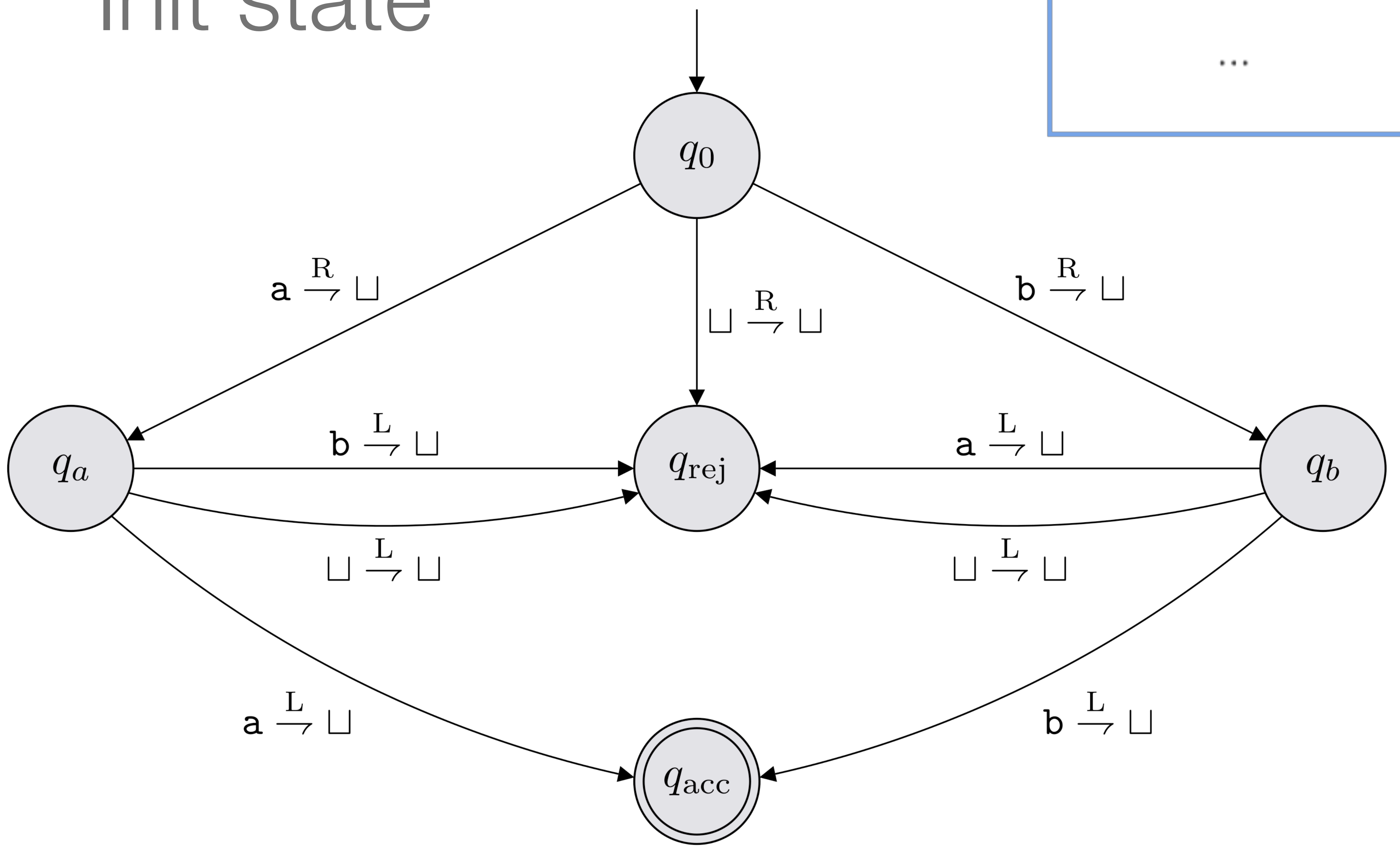
Alphabet

States

Init state

Transitions





Turing Machine - Application

- Complexity Theory (Turing Completeness, etc…)
- Algorithm Design and Analysis
- Computer Design
- Programming Language (BrainF***)

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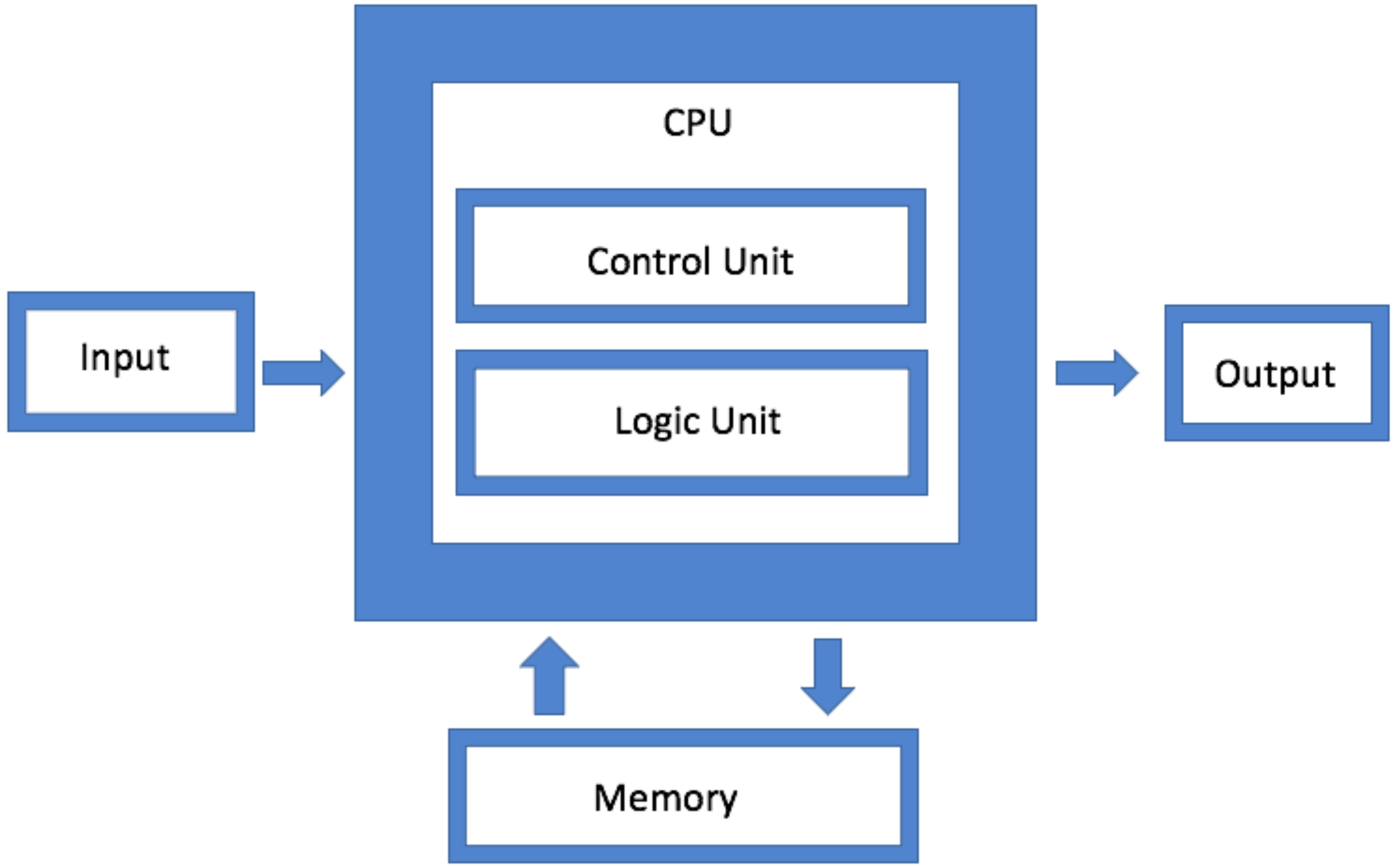
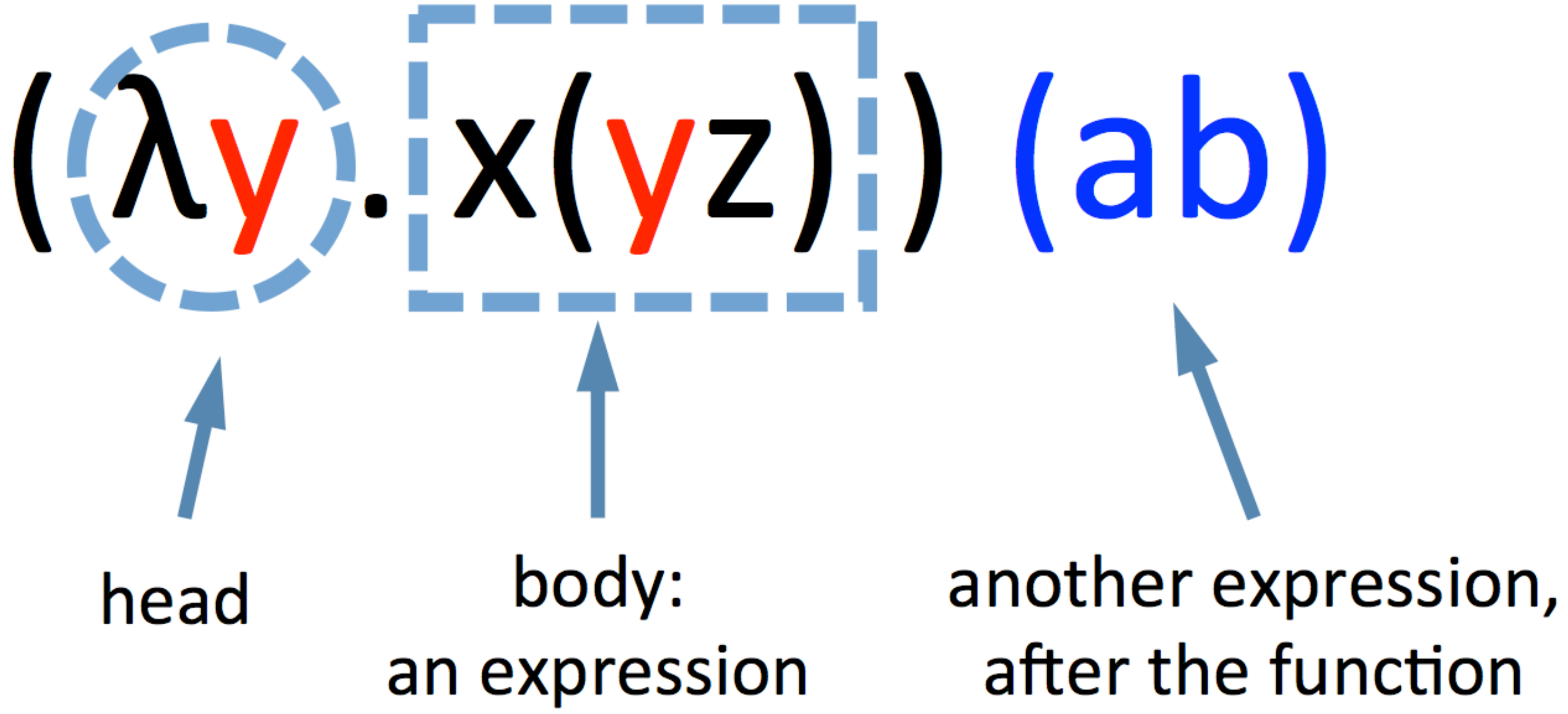
Hello world



Von Neumann Architecture

Church's Lambda Calculus - definition

- $\Sigma = \{v_1, v_2, \cdots\} \cup \{\lambda, .\}$
- Word: sequence of Σ
- Evolution : $(\lambda x . M)N \mapsto M[x := N]$
- Turing Complete



Church’s Lambda Calculus - application

- Programing language Design (System F)
- Logic foundation (Curry-Howard Correspondence)
- Proof Assistant
- Symbolic AI

Probabilistic Turing Machine - Definition

- $TM = \langle \Sigma, Q, q_0, \delta, F \rangle$

Alphabet States Init state Transitions Final states

- $PTM = \langle \Sigma, Q, q_0, \delta_1, \delta_2, F \rangle$

At each step, flip a coin :

- Head : Apply δ_1
- Tail : Apply δ_2

Probabilistic Turing Machine - Application

- Formalize Random Process
- Simply Computation
- New classes (RP, BPP)
- Open problem : P = BPP ?

Quantum Turing Machine - Definition

- Goal : Leverage Superposition and Entanglement

- $QTM = \langle \Sigma, \Gamma, \mathscr{H}, q_0, u, F \rangle$

Tape Alphabet

States

Unitary operation

Hilbert space

- Superposition of configuration, Unitary Evolution, Measurement

Quantum Turing Machine - Key result

- New class : BQP , QMA,
- New Church-Turing-Deutsch Thesis
- Most general model

Quantum Lambda Calculus (Selinger & Valiron, 2009)

- Ad-hoc extension :
  - $\mathbf{new}$ is a function for state preparation.
  - $\mathbf{meas}$ is a function for measurement.
  - a set U of unitary operator
  - “the construction of a concrete model […] is an open problem”

Adiabatic Quantum Computation

- Leverage Adiabatic theorem
- Input : ground state of Hamiltonian
- Program : slow evolution nof Hamiltonian
- Output : Final ground state

Adiabatic Quantum Computation - Application

- Quantum Anneling
- Well-suited to optimization problem
- Physical Implementation : D-wave

Open Problems

- Formal proof of equivalence
- Physical Implementation
- Measure of Fault-Tolerance